Cálculos teóricos:

2 - a) $R_1 = 1{,}31\ [\Omega]$; $L_1 = 0{,}53\ [mH]$

$R_2 = 0{,}32\ [\Omega]$; $L_2 = 0{,}13\ [mH]$

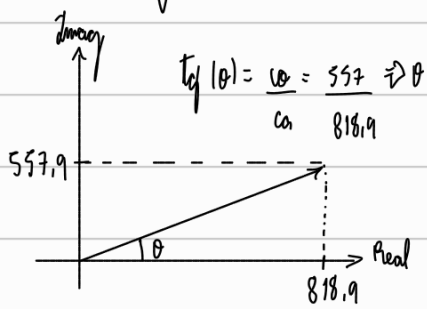
$X_1 = 0{,}203\ [\Omega]$; $X_2 = 0{,}050\ [\Omega]$

$R_m = 1760\ [\Omega]$; $X_m = 1199\ [\Omega]$; $L_m = 3{,}18\ [H]$

$$Z_1 = (R_1 + X_1) = 1{,}31 + 0{,}203j = 1{,}3256 \angle 8{,}80^\circ$$

$$Z_2 = (R_2 + X_2) = 0{,}32 + 0{,}05j = 0{,}323 \angle 8{,}88^\circ$$

$$Z_m = (R_m // X_m) = \frac{1760 \cdot 1199j}{(1760 + 1199j)} = 818{,}9 + 557{,}9j = 990 \angle 34{,}21^\circ$$



$$tg(\theta) = \frac{co}{ca} = \frac{557}{818{,}9} \Rightarrow \theta = arctg(0{,}68) = 34{,}21^\circ$$

Para a carga resistiva: usando 25 [Ω]

$$\dot{V}_{2(L)} = 127 \angle 0^\circ \Rightarrow N = \frac{220}{127} = 1{,}732$$

$$\dot{V}'_{2(L)} = N \cdot V_{2(L)} = 219{,}96 \angle 0^\circ$$

$$\dot{I}_2(L) = \frac{\dot{V}_2(L)}{R_L} = 5{,}08\ [A]$$

$$\dot{I}'_2 = \frac{\dot{I}_{2(L)}}{N} = 2{,}93\ [A]$$

$$\dot{I}_1(L) = \dot{I}'_2(L) + \frac{\dot{V}'_2(L)}{Z_m} = 2{,}93 \angle 0^\circ + \frac{219{,}96 \angle 0^\circ}{990 \angle 34{,}21^\circ} = 3{,}15 \angle -34{,}21^\circ$$

$$\dot{V}_1(L) = \dot{I}_1(L)\, Z_1 + \dot{V}_2(L) = 4{,}17 \angle -25{,}41^\circ + 219{,}96 \angle 0^\circ = 223 \angle -0{,}46^\circ$$

$$\dot{V}'_{2(NL)} = \dot{V}_1(L) \cdot \frac{Z_m}{Z_m + Z_1} = 223 \angle -0{,}46^\circ \cdot \frac{990 \angle 34{,}21^\circ}{990 \angle 34{,}21^\circ + 1{,}31 \angle 8{,}80^\circ} = 222{,}73 \angle -0{,}43^\circ$$

Regulação de tensão:

$$Reg = \frac{V_2'(NL) - V_2'(L)}{V_2'(L)} \cdot 100 = \left(\frac{222,73 - 219,96}{219,96}\right) \cdot 100 = 1,26\%$$

Potências:

$$P_{entrada} = Real\{\dot{V}_1 \cdot \dot{I}_1^*\} = 702,45 \ [W]$$

$$P_{saída} = Real\{\dot{V}_2 \cdot \dot{I}_2^*\} = 652,65 \ [W]$$

$$Perdas = P_{entrada} - P_{saída} = 702,45 - 652,65 = 49,8 \ [W]$$

$$Rendimento \ (\eta) = \frac{P_{saída}}{P_{entrada}} = 92,9\%$$

$$erro = \frac{Calculado - medido}{Calculado}$$

Capacitivo

$I_{2(L)} = \frac{127}{-j28,99} = 4,38 \angle 90°$

$91,5\ \mu F$

$Z_C = -j28,99$

$I'_{2L} = 2,53 \angle 90$

$I_1 = I_2 + \frac{V'_2}{Z_m} = 2,48 \angle 88,15°$

$V_1 = I_1 . Z_1 + V_2$ (219,71)

$0,10 \angle -34,26$

$0,08 - j0,05$

$V'_{2(NL)} = V_1 . \frac{Z_m}{Z_m + Z_1} = 219,23 \angle 0,84$

$3,25 \angle 96,95 + 219,71$

$-0,39 + j3,22 + 219,71 = 219,34 \angle 0,84$

$Reg = \frac{219,23 - 219,71}{219,71} . 100 = -0,2$

$P_{ent} = V_1 . I_1 \cos(\theta_{V_1} - \theta_{I_1}) = 9,59W$ (0,84 − +88,15)

$P_{out} = V_2 . I_2 \cos(\theta_{V_2} - \theta_{I_2}) = 0W$

0 − −90

$$Reg = \frac{V'_{2NL} - V'_{2L}}{V'_{2L}} \cdot 100\% = \frac{220,88 - 219,71}{219,71} \cdot 100\% = 1$$

$$Pent = |V_1| \cdot |I_1| \cos(\underbrace{\overset{-6,96^\circ}{\theta_{V_1}} - \overset{-79,59}{\theta_{I_1}}}_{78,63}) = 128,90W$$

$$221,01 \cdot 2,96 \cdot \underbrace{0,197}$$

$$Pout = |V_2| \cdot |I_2| \cdot \cos(\underbrace{\overset{0}{\theta_{V_2}} - \overset{+80,96}{\theta_{I_2}}}_{0,0006}) = 99,77W$$

$R_1 = 1{,}31\ \Omega$ $\quad R_2 = 0{,}320\ \Omega$ $\quad R_m = 1760\ \Omega$

$X_1 = 0{,}203\ \Omega$ $\quad X_2 = 0{,}050\ \Omega$ $\quad X_m = 1199\ \Omega$

$L_1 = 0{,}53\,mH$ $\quad L_2 = 0{,}13\,mH$ $\quad L_m = 3{,}18\,H$

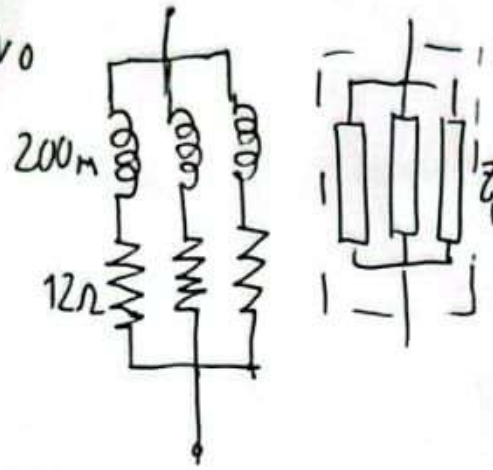
$2130{,}7 \angle 34{,}26°$

$Z_m = 2129{,}6 \angle 34{,}26°$

$Z_1 = 1{,}31 + j\,0{,}203$

$1{,}32 \angle 8{,}8°$

Indutivo



$Z_{Potencial} = 12 + j24\,M\Omega$ $\quad Z_2 = 0{,}32 \angle 8{,}8°$

$Z_{RL} = 4 + j8\,M\Omega$

$$I_2(L) = \frac{V_2(L)}{Z_{RL}} = \frac{127}{25{,}4 \angle 80{,}96°} = 5 \angle -80{,}96°$$

$$I_2' = \frac{I_2}{1{,}73} = 2{,}89 \angle -80{,}96°$$

$$I_1(L) = I_2' + \frac{V_2'}{Z_m} = 2{,}89 \angle -80{,}96 + \frac{\overbrace{127 \cdot 1{,}73}^{219{,}71}}{2129{,}6 \angle 34{,}26°}$$

$$I_1 = 2{,}89 \angle -80{,}96 + 0{,}103 \angle -34{,}26°$$

$$I_1 = 0{,}45 - j\,2{,}85 + 0{,}085 - j\,0{,}058$$

$$I_1 = 0{,}535 - j\,2{,}908$$

$$I_1 = 2{,}96 \angle -79{,}58°$$

$$V_{1L} = \underbrace{I_1 \cdot Z_1}_{2{,}96 \angle -79{,}58 \;\; 1{,}33 \angle 8{,}8°} + \underbrace{V_2'}_{219{,}71} =$$

$$V_{2NL}' = V_{1L} \cdot \frac{Z_m}{Z_m + Z_1} = 220{,}88$$

$$3{,}94 \angle -70{,}78 + 219{,}71 \quad 221{,}01$$

$$\boxed{V_{1L} = 221{,}01 \angle -0{,}96°}$$

$$V_{1L} = -j\,3{,}90 - j\,3{,}72 + 219{,}71 = -j\,3{,}72 =$$